

1. Data Split

Subjects are randomly split into base model subjects (80 subjects) and fine-tuning subjects (all remaining subjects)

The base model data is then split into training and validation sets (current 80/20)

2. Data Augmentation

Subject training data for the base model is augmented

- For each augmented trial, it is randomly decided whether 1 or 2 augmentations will be applied from the below list
- Which augmentations are selected is also random

Augmentations:

- Addition of Guassian noise
- Random amplitude scaling (currently 0.9 to 1.1)
- Perturb specific frequency bands without time-domain distortion
- Application of smooth warping
- Combination with another un-augmented trial

Validation data is not augmented

Normalization is calculated based on training data and then applied to training and validation data sets

3. Feature Extraction

Fit CSP (currently 6 components) on original un-augmented data and apply transformation to both original and augmented data

Calculate average band power for 8-12 Hz and 12-30 Hz

Normalize features

4. Model

Note: dropout used throughout

In order:

CNN Encoder

- Block 1
 - Temporal convolution (16x, 1 x 64 kernel)
 - Batch normalization
 - Spatial convolution (32x, n_channels x 1 kernel, per channel)
 - Batch normalization
 - ELU activation
 - Average pooling (4 to 1)
- Block 2
 - Depthwise Convolution (32x, 1 x 16 kernel, per channel)
 - Pointwise Convolution (32x, 1 x 1 kernel)
 - Batch normalization
 - ELU activation
 - Average pooling (8 to 1)
- Final transformation
 - Converts to embedding dimension (64)

Combines CNN output with CSP and frequency features

- CSP encoder and frequency encoder both include layer normalization and GELU activation

Positional encoding adds sinusoidal positional information to each token

Multi-Head Self-Attention

- 4 attention heads
- 2 layers

- 64 dimensional embedding

Feed-forward network

- Two dense layers
- 64 --> 256 --> 64
- GELU activation

Each transformer block uses:

- Two residual connections (one after attention, one after FF)
- Two layer normalization operations

Average pooling over time dimension

Two layer MLP for classification

- 64 --> 32
- GELU activation
- 32 --> 2

5. Base Model Training Process

- AdamW optimizer (learning rate = 0.0004)
- Batch size = 24
- Uses learning rate scheduler

6. Fine Tuning

Uses a single subject's data to attempt to classify their trials
Currently uses 50/50 split for validation data so there is enough

- Loads best base model (chosen by validation accuracy)
- Freezes all non-classification layers
- Uses lower learning rate
- Normalizes based on single subjects training data

- Currently not very successful, needs adjustments

Have attempted using data augmentation here but currently doesn't improve it over no augmentation